DAY 32 | PRE-READ

# Quality Checks Before Every Release

A light read before Day 32. About 15 minutes, no code required.

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## Why read this?

Yesterday you measured how good an AI system is. Today you make those measurements automatic, so that every change to a prompt, model or setting is checked for quality and cost before it reaches users.

## 1. The answer key

Teachers mark exams against an answer key that experts have checked. A **golden dataset** is the answer key for an AI system: questions with verified answers, kept safe and updated when policies change.

## 2. Small changes, surprising effects

Changing one word in a prompt, or switching to a cheaper model to save money, can quietly break answers that worked yesterday. **Regression testing** reruns the golden questions after every change and flags anything that got worse, just like the tests you wrote for code.

**Key idea:** treat prompts like code. Version them, test them, and review changes.

## 3. Did we fetch the right pages?

| Measure | Question | Everyday version |
|---|---|---|
| Precision | How much of what we fetched was useful? | Of 10 books brought, how many were relevant? |
| Recall | How much of what we needed did we fetch? | Of the 3 books needed, how many arrived? |

## 4. Pressing the right button NEW (v4)

When an AI agent takes an action, the action is the answer. If a customer asks for a refund of 300 on order 77, the agent must call the refund tool with exactly that order and amount. **Tool-call fidelity** tests check this the way a bank checks every transfer.

## 5. Good and affordable NEW (v4)

A change that makes answers slightly better but triples the bill is not automatically a win. Quality gates now also check **tokens and cost per question**, so nobody is surprised by the invoice.

## 6. Think about these before class

- Who in a telecom company could verify golden answers for a support assistant?
- What change might someone make to save cost that could quietly reduce quality?
- Can you think of two correct answers to a question that share almost no words?

Kihon (基本): the foundation, the basics